Aim:Performance Evaluation of MANET Protocols
Measure throughput, packet delivery ratio, and end-to-end delay for different MANET routing protocols.
CODE:

```
# MANET_AODV.tcl

set val(chan)          Channel/WirelessChannel
set val(prop)          Propagation/TwoRayGround
set val(netif)         Phy/WirelessPhy
set val(mac)           Mac/802_11
set val(ifq)           Queue/DropTail/PriQueue
set val(ifqlen)        50
set val(ll)            LL
set val(ant)           Antenna/OmniAntenna
set val(nn)            5
set val(rp)            AODV
set val(x)             500
set val(y)             500
set val(stop)          20.0

set ns [new Simulator]

set tracefd [open manet.tr w]
$ns trace-all $tracefd

set namfd [open manet.nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)

set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)

create-god $val(nn)

set chan_1_ [new $val(chan)]

$ns node-config \
   -adhocRouting $val(rp) \
   -llType $val(ll) \
   -macType $val(mac) \
   -ifqType $val(ifq) \
   -ifqLen $val(ifqlen) \
   -antType $val(ant) \
   -propType $val(prop) \
   -phyType $val(netif) \
   -channel $chan_1_ \
   -topoInstance $topo \
   -agentTrace ON \
   -routerTrace ON \
   -macTrace ON \
```

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```
   -movementTrace ON

for {set i 0} {$i < $val(nn)} {incr i} {
   set node_(i)[ns node]
}

$node_(0) set X_ 50
$node_(0) set Y_ 100

$node_(1) set X_ 150
$node_(1) set Y_ 100

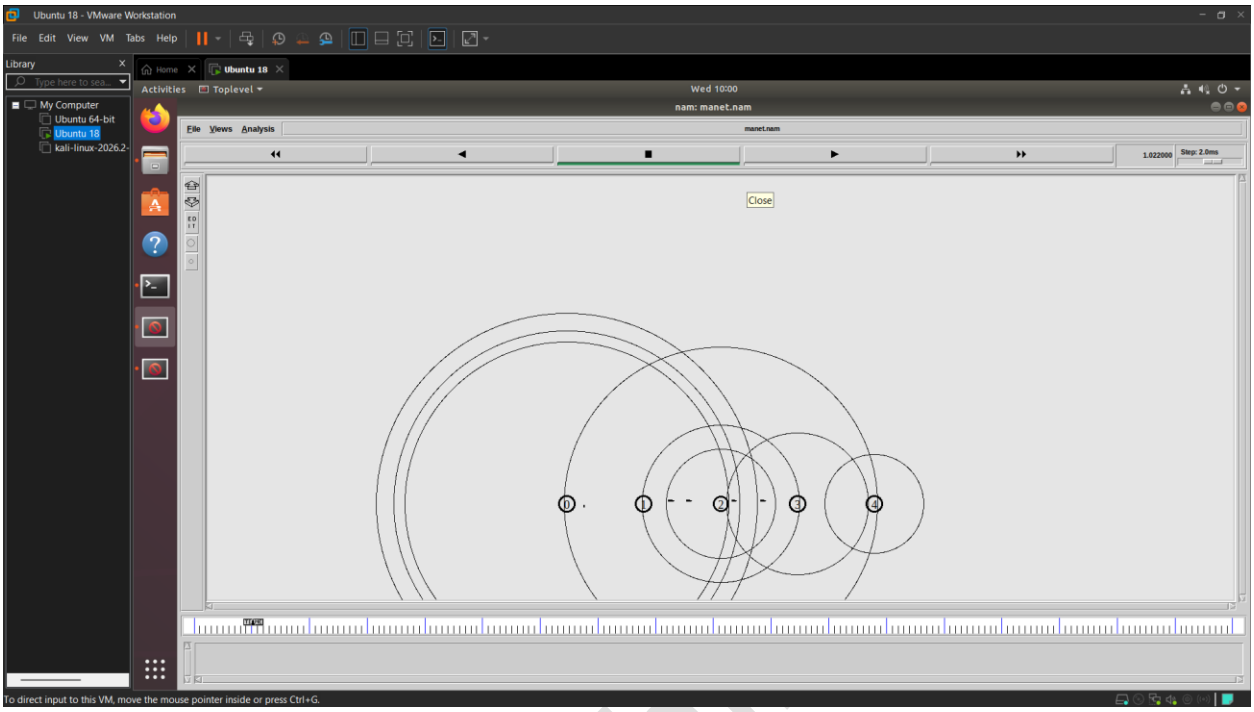
$node_(2) set X_ 250
$node_(2) set Y_ 100

$node_(3) set X_ 350
$node_(3) set Y_ 100

$node_(4) set X_ 450
$node_(4) set Y_ 100
for {set i 0} {$i < $val(nn)} {incr i} {
   node_(i) set Z_ 0
   $ns initial_node_pos node_(i) 20
}
# UDP/CBR Traffic
set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0
set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0
$ns connect $udp0 $null0
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.05
$cbr0 attach-agent $udp0
nsat1.0"cbr0 start"
nsat18.0"cbr0 stop"
proc finish {} {
   global ns tracefd namfd
   $ns flush-trace
   close $tracefd
   close $namfd
   puts "Simulation Finished"
   exec nam manet.nam &
   exit 0
}
$ns at $val(stop) "finish"
puts "Starting Simulation..."
$ns run
```

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OUTPUT:



Observation

- Five wireless nodes were created in the simulation area.
- Node 0 acted as the source and Node 4 as the destination.
- UDP was used as the transport protocol.
- CBR traffic generated continuous packet transmission.
- AODV established routes dynamically between nodes.
- Packet delivery was successfully visualized in NAM.

Result

The MANET network was successfully implemented using the AODV routing protocol. Data packets were transmitted from the source node to the destination node, and the communication process was observed through NAM animation.

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